Vaibhav Sharma

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Summary

Backend and systems engineering student specializing in event-driven microservices, enterprise workflow backends, and secure Linux-level programming. Open-sourced Arachnode — a distributed job-discovery pipeline (Redis Streams, FastAPI, Docker) with **26 stars & 47 forks** — and serves as Project Admin mentoring GSSoC contributors. Built production-oriented systems spanning distributed pipelines, constraint-based scheduling, multi-role approval workflows, QR-based gate management, and AI model orchestration. Strong in API design, transactional data modeling, RBAC, and multi-service integration. Graduating 2027. **Targeting SDE-1 / Junior Backend Engineer roles.**

Education

Mangalore Institute of Technology and Engineering — Sep 2023 – Jun 2027
B.E. Computer Science and Engineering | ***CGPA: 9.43 / 10*** — *Karnataka, India*

Technical Skills

Languages: Java, Python, SQL, C, JavaScript (Node.js)
Backend: Spring Boot, Spring Security, Spring Data JPA, FastAPI, Flask, Express.js, REST API Design, JWT, RBAC
Databases & Messaging: PostgreSQL, Redis Streams, MySQL, MongoDB, Firestore, Flyway, Prisma ORM
DevOps & Tooling: Docker, Docker Compose, Nginx, GitHub Actions, Linux, Bash, Git, Maven
Testing: pytest, testcontainers, H2 (integration), Unit / Contract / e2e
Architecture: Event-Driven Microservices, Consumer Groups, Modular Monolith, API Gateway, State Machines, Audit Trails, Constraint Scheduling
Systems: Linux Namespaces, Seccomp, rlimits, llama.cpp, Syscall Analysis, .NET (bridge layer)
ML / CV: PyTorch, TensorFlow, MusicGen, YAMNet, MediaPipe, OpenCV, Ollama, Model Serving, Inference Optimization
Frontend: React, Vite, Flutter, Dart

Experience

Backend Engineering Intern — Jun 2026 – Present
LazyStudents.in — *Remote*
- Primary backend engineer on a plagiarism detection and reduction service — designing the text-comparison pipeline, backend API, and suggestion engine from scratch as the sole engineer on the feature.
- Own end-to-end quality of **40+ platform tools**: writing structured test plans, identifying integration failures, and filing reproducible bug reports consumed directly by the backend team for prioritization and hotfixes.
- Leading the interns team, coordinating task allocation, review cycles, and delivery timelines in an Agile environment.
- Next assignment: building an AI-based workspace product — full backend ownership from design through delivery.

AI/ML Backend Intern — Sep 2025 – Nov 2025
Infosys Springboard — *Virtual*
- Orchestrated 3 heterogeneous ML models (MusicGen audio generation, fine-tuned GPT lyric model, YAMNet + GTZAN genre classifier) behind a Flask API layer for MoodHarmonics, an end-to-end AI music generation platform.
- Implemented model preloading at server startup and graceful fallback routing (local inference preferred, Hugging Face fallback), cutting average initialization latency from ∼2.2s to ∼1.2s (**45% reduction**), measured over 500+ sequential requests.
- Designed error-handling middleware ensuring inference continuity under model failures; persisted semi-structured song metadata and hashed user credentials in MongoDB.
- Coordinated frontend–backend interactions under real CORS and latency constraints; enforced lyric structure via fine-tuning over base GPT-2 (which produced descriptive text, not lyrical verse/chorus output).

Flutter Developer Intern — Jul 2025 – Sep 2025
ClubChat — *Remote*
- Shipped event-feed and real-time chat features to a production mobile application with **8,000+ users**.
- Resolved **18 production stability issues** across two releases — including 3 critical crash-on-launch regressions — through systematic log analysis, root-cause isolation, and device-level reproduction.
- Operated inside sprint-based release cycles collaborating with backend engineers and product designers.

Projects

Arachnode | *Python, FastAPI, Redis Streams, PostgreSQL, Docker, Scrapy, Playwright, pytest, testcontainers* | 26 stars · 47 forks · 143 c
- Architected a self-hosted event-driven job-discovery platform across **7 independent microservices** (crawler, platform scraper, aggregator, contact discovery, email generator, APScheduler orchestrator, API gateway) communicating via Redis Streams consumer groups — chose Redis Streams over Kafka for sufficient at-least-once delivery semantics with significantly lower operational overhead for a self-hosted single-node deployment.
- Implemented a **4-tier test suite**: unit tests against mock objects; integration tests using `testcontainers` (ephemeral Postgres + Redis spun per test run); contract tests asserting API schema consistency across decoupled services; e2e tests traversing full dashboard flows.
- Built OSINT-based recruiter contact enrichment (GitHub API + SMTP domain verification) and Ollama-powered cold-email drafting via Jinja2 templates; full discovery-to-draft cycle automated via APScheduler cron jobs, eliminating hours of daily manual job-search effort.
- Serves as Project Admin: reviewing **25+ PRs**, enforcing API contract standards, managing CI/GitHub Actions pipelines, coordinating releases, and mentoring GSSoC / ELUSOC contributors across an active open-source community.

HPMS — Hostel Permission Management System | *Node.js, Express, PostgreSQL, Prisma, React, JWT, Docker, Nginx, QR Codes* 20

- Designed and built a full digital replacement for the college hostel's paper-based permission-slip process: **5-role workflow** (Student / Warden / Chief Warden / Admin / Security) with JWT-signed QR certificates generated on approval — valid for exactly 2 scans (exit + entry), auto-invalidated on completion.
- Modular monolith backend (Node.js + Express + Prisma) across 7 feature modules (auth, users, master, requests, certificates, gate, admin) served behind Nginx reverse proxy with SSL termination — required for `getUserMedia` camera access in the gate scanner web app (`html5-qrcode`).
- node-cron scheduled job detecting overdue student returns and triggering warden alerts; SMTP notification pipeline to parents on request approval, exit scan, and return scan (with warden contact included per SRS requirement).
- Attempted Tatvik TMF20 biometric integration: built a .NET 4.8 bridge (TMF20Bridge) exposing the proprietary SDK over HTTP and a .NET 8 FingerprintService calling it, storing fingerprint templates in Firestore — hit a firmware wall (SDK designed to block third-party callers); pivoted to the QR-based flow as the production-viable solution.
- Delivered a full SRS (40+ functional requirements, ERD, 4 system process flow diagrams); proposed to college administration for institutional deployment.

Appraisal Management System | *Java 21, Spring Boot 4, Spring Security, PostgreSQL, Flyway, JWT, Docker, OpenAPI* 2026

- Built a state-machine-driven enterprise workflow backend replacing a paper-based faculty appraisal process: `DRAFT → SUBMITTED → HOD_REVIEWED → FINALIZED → LOCKED`, with legal revert paths and immutability enforcement post-finalization.
- Full audit trail on every state transition (userId, timestamp, old/new value); JWT authentication with RBAC across 4 roles (Faculty / HOD / Principal / Admin); configurable scoring weights with server-side validation that weights sum to exactly 100%.
- 13 Flyway migrations applied cleanly; `@Valid` request validation on all controller endpoints; env-injected credentials via Docker Compose with no hardcoded secrets in image; Swagger/OpenAPI documentation auto-generated.
- Full API surface: auth, faculty lifecycle, appraisal submission, reviewer workflows (approve/reject/revert), cycle management, form structure configuration, scoring weight management, and export endpoints.

Academic Timetable Scheduling Engine | *Java 17, Spring Boot 3, PostgreSQL, Flyway, H2, Apache POI, OpenPDF, Docker* 2026

- Greedy constraint-propagation scheduling engine enforcing **8 hard constraints**: no double-booking across faculty/room/section, 55-minute consecutive-class gap enforcement, designation-based load limits (maxLectureHours, maxLabHours, maxTotalHours), room-type matching (LAB vs CLASSROOM), break slot isolation, Saturday post-lunch exclusion.
- Section-first generation strategy prevents partial slot starvation — each section fully secures its lab slots before the next begins. Fallback faculty assignment iterates priority-ranked `FacultyPreference` entries before failing.
- Semester parity partitioning (ODD: 1,3,5,7 / EVEN: 2,4,6,8) enables half-semester regeneration without disturbing published schedules. Drag-and-drop slot update endpoint re-validates all constraints before committing.
- 14-entity normalized domain model; Flyway V1–V10; H2-backed integration tests validating constraint correctness; Apache POI Excel and OpenPDF export; two-stage Docker build with env-injected credentials; generates conflict-free timetables in **under 2 minutes** vs. days of manual scheduling.

Offline AI Assistant Shell | *C, llama.cpp, TinyLlama-1.1B, Seccomp, Linux Namespaces, rlimits, CMake* 2025

- Built a sandboxed C shell integrating TinyLlama-1.1B via `llama.cpp` for offline AI-assisted command guidance — fully air-gapped, no internet connection required post-setup.
- Four-layer security model: Seccomp syscall whitelist (blocking unsafe system calls), PID/mount namespace isolation (containing the AI worker process), capability dropping (preventing privilege escalation), rlimit caps (bounding CPU and memory usage).
- First-query latency ∼2–3s (model load); subsequent queries ∼1–2s reusing context; 10–20 tok/s on CPU at ∼1–1.5GB memory. Functional test suite (`test_assist.sh`), MD5 model integrity verification, stress test script, and `strace`-based syscall audit workflow.

MoodHarmonics | *Python, Flask, MusicGen, YAMNet, GPT (fine-tuned), PyTorch, TensorFlow, MongoDB, React* 2025

- Orchestrated 3 heterogeneous ML models with different runtimes and constraints behind a single Flask API: Meta MusicGen (text-to-audio), a fine-tuned GPT lyric model (verse/chorus structure enforcement), and YAMNet + GTZAN classifier (genre + confidence score for automatic categorization).
- Model preloading at server startup + graceful local-first / Hugging Face fallback paths to manage heavy initialization (∼seconds) and model failure modes; error-handling middleware ensuring continuity across all 3 inference paths.
- Semi-structured song metadata and hashed user credentials stored in MongoDB; generated audio served from local file storage; frontend–backend CORS and latency constraints handled under real performance pressure.

Air Notepad | *Python, OpenCV, MediaPipe, NumPy, Jupyter* 2024

- Real-time gesture-based drawing application tracking **21 hand landmarks per hand** via MediaPipe Hand Landmarker, mapping raw landmark coordinates to drawing strokes via weighted moving-average smoothing (configurable factor, default 0.3) to reduce jitter.
- Dual-hand control: left hand for tool/color selection via pinch gesture (Euclidean distance threshold between thumb and index); right hand for drawing via index-finger extension detection using relative Y-coordinates.
- HD canvas (1280×720), 3 drawing tools (Pen 3px / Brush 8px / Eraser 20px), 8-color palette; detection and tracking confidence thresholds at 0.8; NumPy array operations for minimal per-frame latency. LinkedIn post reached **176,000+ impressions**.

Certifications

Oracle (4 certs): OCI 2025 DevOps Professional (Nov 2025) · OCI 2025 Data Science Professional (Oct 2025) · OCI 2025 Generative AI Professional (Oct 2025) · OCI 2025 Generative AI Foundations Associate (Sep 2025)

Google Cloud / Coursera (3 certs): Cloud Security Engineer Specialization (Jul 2024) · Security in Google Cloud Specialization (Sep 2024) · Google Cybersecurity Professional Certificate

Leadership & Open Source

Project Admin — Arachnode | GSSoC / ELUSOC / Aperture 3.0 Feb 2026 – Present

- Reviewing and merging contributor PRs, enforcing API contract standards, maintaining CI/GitHub Actions pipelines, and coordinating releases for an active open-source project with **26 stars and 47 forks**.
- Mentoring first-time open-source contributors from GSSoC and ELUSOC programs on backend API design and Python best practices.

- Merged 2 PRs into the GitCanvas project under Aperture 3.0 after maintainer review.

President, Coders Club — MITE Oct 2025 – Present

- Organized **Hack n' Seek**, a hybrid technical event drawing **700+ online and 180 on-site participants** across multiple colleges.
- Delivered 6 technical workshops on Git internals, Data Structures, and backend architecture to **60+ students**.
- Active engagement with developer communities including GDG, cloud, AI, and cybersecurity groups.